Software Engineering

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

PROBLEM STATEMENT #42: Internal Microservice Catalog & Health Portal

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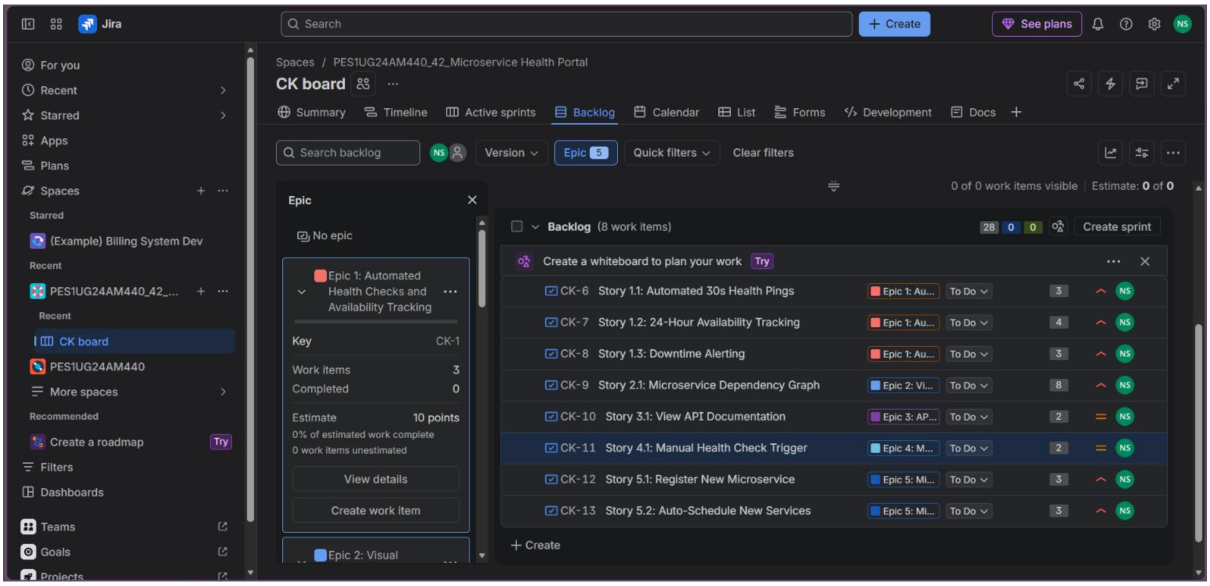
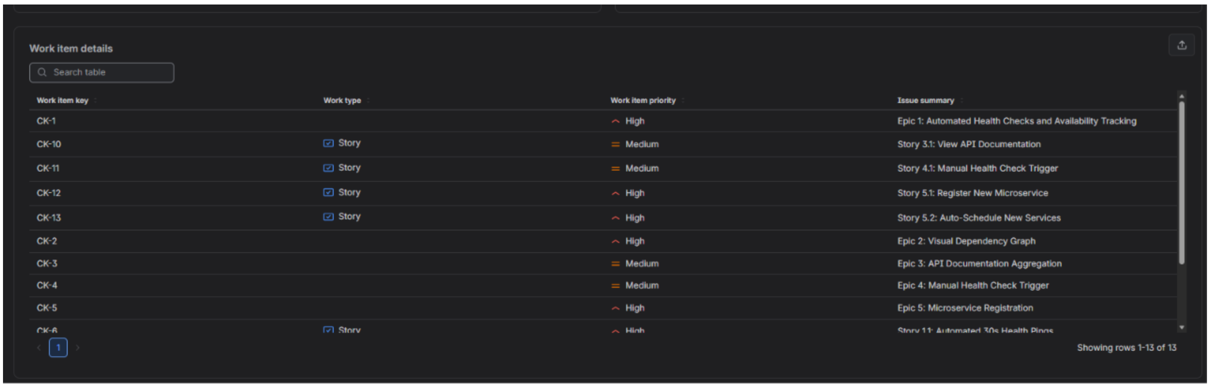
Section: H

Jira backlog with Epics and User Stories and Story point assignments:

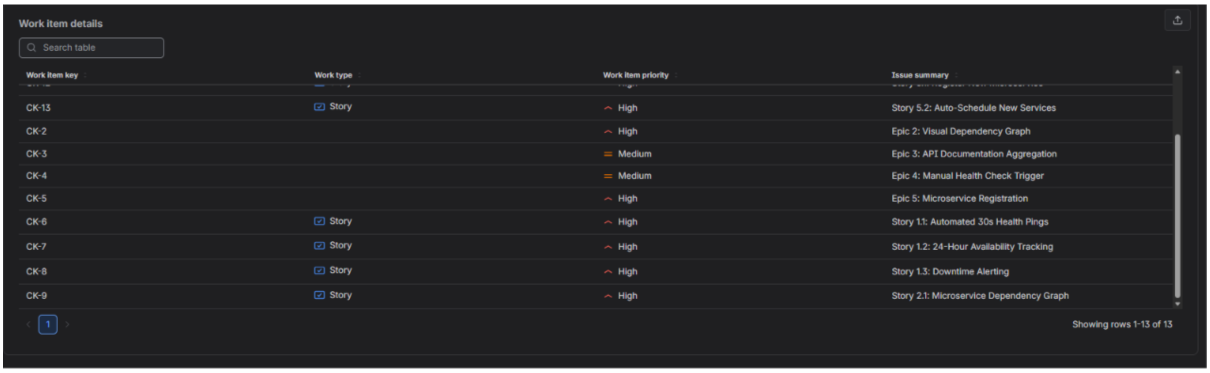




Work item details

Work item key	Work type	Work item priority	Issue summary
CK-1		High	Epic 1: Automated Health Checks and Availability Tracking
CK-10	Story	Medium	Story 3.1: View API Documentation
CK-11	Story	Medium	Story 4.1: Manual Health Check Trigger
CK-12	Story	High	Story 5.1: Register New Microservice
CK-13	Story	High	Story 5.2: Auto-Schedule New Services
CK-2		High	Epic 2: Visual Dependency Graph
CK-3		Medium	Epic 3: API Documentation Aggregation
CK-4		Medium	Epic 4: Manual Health Check Trigger
CK-5		High	Epic 5: Microservice Registration
CK-6	Story	High	Story 1.1: Automated 30s Health Pings

Showing rows 1-13 of 13



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CK-2		High	Epic 2: Visual Dependency Graph
CK-3		Medium	Epic 3: API Documentation Aggregation
CK-4		Medium	Epic 4: Manual Health Check Trigger
CK-5		High	Epic 5: Microservice Registration
CK-6	Story	High	Story 1.1: Automated 30s Health Pings
CK-7	Story	High	Story 1.2: 24-Hour Availability Tracking
CK-8	Story	High	Story 1.3: Downtime Alerting
CK-9	Story	High	Story 2.1: Microservice Dependency Graph

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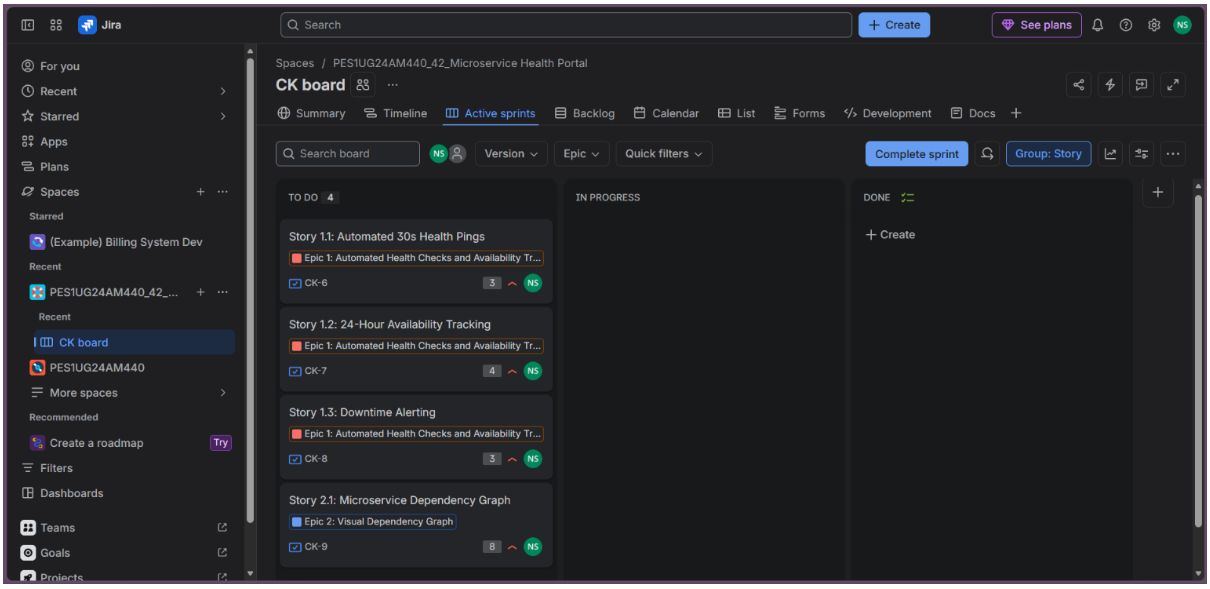
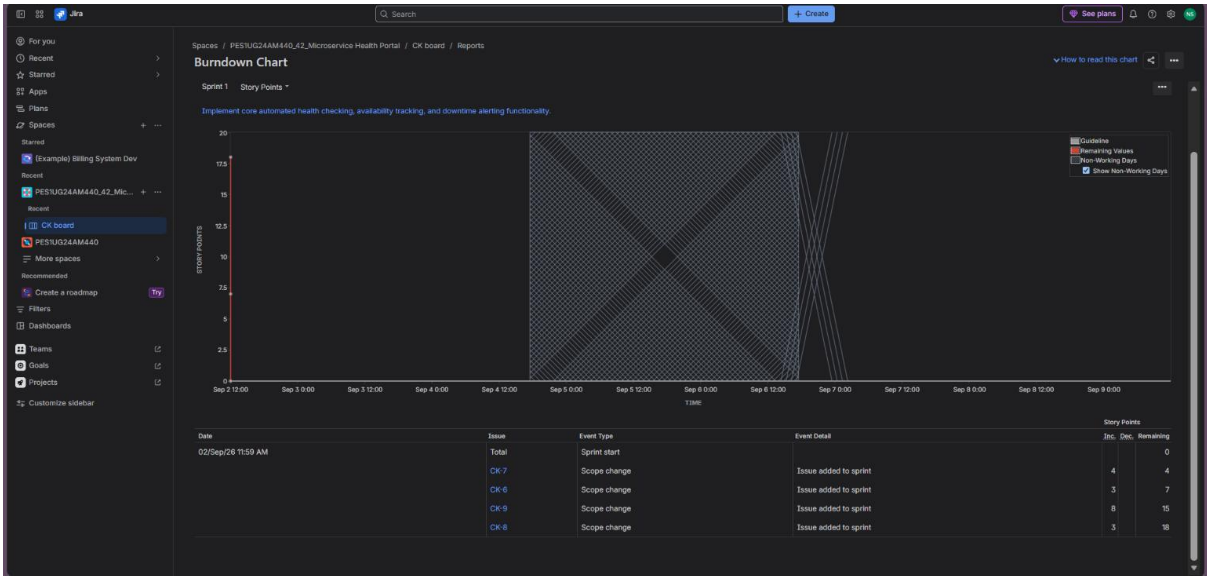
Sprint board (Active Sprint view):



Burndown Chart:



Date	Issue	Event Type	Event Detail	Story Points Inc.	Story Points Dec.	Story Points Remaining
02/Sep/26 11:59 AM	Total	Sprint start				0
	CK-7	Scope change	Issue added to sprint	4		4
	CK-6	Scope change	Issue added to sprint	3		7
	CK-9	Scope change	Issue added to sprint	8		15
	CK-8	Scope change	Issue added to sprint	3		18

Reflection Questions:

1. Did your estimations reflect the actual effort?
   No, the estimations did not reflect actual execution effort because tasks assigned higher Fibonacci story points completed instantaneously during the lab simulation rather than taking their projected multi-day timeline.

2. Was your backlog well-prioritized?
   Yes, the backlog was effectively prioritized by aligning high-impact functional requirements-such as automated health checks and critical downtime alerts-to a high-priority status, ensuring essential system reliability tasks were tackled first.

3. How did your simulated sprint align with your plan?
   The simulated sprint moved faster than a traditional execution plan, allowing tasks to cross from the backlog into active status and completion rapidly to satisfy the 60-minute timeframe of the laboratory exercise.

4. What insights did the burndown chart give about your team’s capacity?
   The burndown chart exhibited a sharp, immediate vertical drop on the first day, indicating that work was resolved instantaneously rather than following a linear daily velocity line due to the nature of the lab walkthrough.